



INFRASTRUCTURE & PLATFORM SPEND

Operating Cost Projection

Three scaling scenarios — 5,000 / 50,000 / 500,000 users — based on the current stack: Next.js on Vercel, Supabase, Stripe Connect, Sentry, web push & FCM.

TIER 1

5,000 users

~\$80–110/mo

platform only

TIER 2

50,000 users

~\$600–1,200/mo

platform only

TIER 3

500,000 users

~\$5k–14k/mo

platform only

Forecast **Assumptions**

Costs are direction-of-magnitude and assume the stack documented in CLAUDE.md. All figures exclude VAT.

MODELLING BASELINE

- **Activity:** ~30% of registered users are monthly-active
- **Bookings:** 2 bookings per active user per month on average
- **Booking fee:** £25–£35 paid via wallet top-up → escrow → Connect transfer
- **Push fan-out:** ~80% of users with a valid web or FCM subscription
- **Sentry sampling:** 10% traces (per instrumentation-client.ts)
- **Stripe model:** 2.9% + 30p Checkout, \$2/month Connect MAU, £0.20/transfer

How to read this report

Each tier separates **platform fixed cost** (what you write a cheque for) from **Stripe pass-through** (variable, taken from GMV before payout). Stripe pass-through grows linearly with revenue, so it doesn't determine whether the business is solvent — but Connect's \$2/active-payee fee does deserve attention at the high tier.

5K	Tier 1 — 5,000 users ~3,000 MAU · ~6,000 bookings/mo · ~£180k GMV/mo	PLATFORM FIXED ~\$80–110
Service	Plan / notes	Monthly cost
Vercel	Pro (required — Hobby throttles cron to 1/day per CLAUDE.md)	\$20 + ~\$10
Supabase	Pro — 8GB DB, 250GB egress, 100GB storage	\$25
Sentry	Team plan, EU region	\$26
Resend	Free tier up to 3k/mo, Pro if exceeded	\$0–\$20
Mapbox	Free tier (≤50k map loads/mo)	\$0
FCM / Web Push	Self-hosted VAPID + Firebase free quota	\$0
Domain & misc	whistleconnect.co.uk	~\$2
Platform total	excluding Stripe pass-through	~\$80–110/mo
Stripe pass-through	2.9% + 30p × ~£180k GMV + Connect MAU fees	~£5.5k from GMV

Headroom: Current architecture is comfortably within free-tier headroom on almost every service. Margin is healthy; focus engineering time on the Phase 1 security hardening backlog, not infra.

50K**Tier 2 — 50,000 users**

~30,000 MAU · ~60,000 bookings/mo · ~£1.8M GMV/mo

PLATFORM FIXED

~\$600–1,200

Service	Plan / notes	Monthly cost
Vercel	Pro + function/bandwidth overage (push fan-out is hot path)	\$170–\$320
Supabase	Pro + Small compute add-on, egress overage, possible read replica	\$200–\$400
Sentry	Team + event overage, likely Business tier	\$80–\$250
Resend	Pro / Scale (booking + dispute emails)	\$90
Mapbox	Paid map loads (~150–300k)	\$50–\$150
FCM	Still within free quota	\$0
Platform total	excluding Stripe pass-through	~\$600–1,200/mo
Stripe pass-through	~£1.8M GMV + ~10k active Connect payees	~£55k + ~\$20k MAU

Tipping points: The RLS gaps in `0002_rls_policies.sql` and the `getThreads` N+1 stop being code smells and start driving real egress + compute spend. Resolve before crossing 30k MAU.

High-Scale Tier

At this scale the current architecture needs work before pure scale-up makes sense. The numbers below assume escrow-release has been moved to Supabase pg_cron and push fan-out runs via a managed queue (Inngest / QStash / Trigger.dev).

500K	Tier 3 — 500,000 users ~300,000 MAU · ~600,000 bookings/mo · ~£18M GMV/mo		PLATFORM FIXED
			~\$5k–14k
Service	Plan / notes	Monthly cost	
Vercel	Enterprise — or heavy Pro + Edge Functions	\$2,000–\$5,000	
Supabase	Team / Enterprise + Large compute + read replica + PITR	\$1,000–\$3,000	
Queue (Inngest / QStash)	Push fan-out + retry; required at this scale	\$200–\$500	
Sentry	Business + heavy event overage	\$500–\$1,500	
Resend	Scale tier	\$400–\$900	
Mapbox	~1.5–3M map loads/mo	\$500–\$2,000	
FCM	Still free at this volume	\$0	
CDN / image opt overage	Avatar + venue photos	\$200–\$500	
Platform total	excluding Stripe pass-through	~\$5k–14k/mo	
Stripe pass-through	~£18M GMV + ~100k Connect payees	~£550k + ~\$200k MAU	

Hidden line item: Stripe Connect Express is **\$2 / active payout account / month**. At ~100k active refs that's ~\$200k/year regardless of GMV — independent of card fees.

Architectural cost drivers

5k → 50k

RLS holes + getThreads N+1 (both flagged in CLAUDE.md) become measurable spend, not just tech debt. Fix before scaling.

50k → 500k

Vercel cron + serverless can't run the push fan-out efficiently. Move escrow-release to pg_cron; route notifications through a real queue.

Notification fan-out

A single POST `/api/admin/broadcast-push` already touches every `push_subscription` row. At 500k users this needs paginated dispatch + retry.

Stripe Connect MAU

\$2/active-payee is the largest non-card Stripe line. Aggregate payouts (weekly cycles) to reduce surprise charges from low-volume refs.

Disclaimer. Figures are forward-looking projections drawn from public pricing pages for Vercel, Supabase, Sentry, Resend, Mapbox, Stripe and Firebase. Actual costs depend on push opt-in rate, average bookings per user, regional egress, plan negotiation and engineering choices made before crossing each tier. Stripe pass-through figures are taken from GMV before payout and don't appear on a platform invoice — they reduce gross margin.